

Nada Ashraf

AI Engineer | ML engineer

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🔗 nada-ai-engineer-portfolio.vercel.app 🏠 GitHub

Summary

Computer Science graduate specializing in Artificial Intelligence with hands-on experience building end-to-end AI applications across Machine Learning, Deep Learning, NLP, LLMs, RAG, and Agentic AI. Experienced in developing AI systems from data preprocessing and model experimentation to retrieval pipelines, agent workflows, API development, and interactive applications. Proficient in Python, PyTorch, TensorFlow, FastAPI, LangGraph, Hugging Face, and modern LLM application development.

Technical Skills

Programming Languages:

Python, C++, Java, SQL

AI & ML: Machine Learning, Deep Learning, NLP, Computer Vision, Recommendation Systems, Generative AI, LLMs, Model Evaluation

LLM & Agentic AI: RAG, Prompt Engineering, LLM Fine-Tuning, QLoRA, PEFT, LangGraph, Embeddings, Vector Databases

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, Unsloth, Scikit-learn, OpenCV, Pandas, NumPy, FastAPI, Streamlit

Engineering & Deployment: REST APIs, Docker, Git, GitHub, GitHub Actions, CI/CD

Databases & Cloud: SQL, NoSQL, Vector Databases, Azure

Experience

AI intern- Inspire for Solutions Development 🔗

08/2026

- Developed an AI-powered **Leave Request Evaluation System** using **FastAPI, RAG, LangGraph, embeddings, and Pinecone** to evaluate employee leave requests against company policy.
- Built a policy ingestion and retrieval pipeline to retrieve relevant policy sections and generate **citation-grounded evaluations** for approval, denial, or review-required cases.
- Designed an agentic workflow to validate requests, determine attachment requirements, analyze supporting documents, and produce policy-grounded decisions.
- Developed a **Simple AI Digital Twin** using **LangGraph and Streamlit**, enabling users to define goals, preferences, and tasks through an interactive interface.
- Implemented agent tools for retrieving, creating, and updating tasks, allowing the agent to recommend the next action based on task priority and deadlines.
- Integrated in-memory context into the agent workflow to maintain relevant user and task information during interactions.

Machine Learning/AI training 🔗 , python, AI , ML, EDA

07/2025 – 08/2025

Built an **NLP-based chatbot system** that interprets user queries and generates relevant responses using preprocessed datasets, integrated into a real-time interactive interface using Streamlit.

- Gained hands-on experience in core **Machine Learning concepts**, including supervised learning models and their real-world applications.
- Performed **Exploratory Data Analysis (EDA)** to understand datasets, identify patterns, and support data-driven decision-making.
- Applied **data preprocessing techniques** such as data cleaning, handling missing values, encoding, and feature scaling to prepare datasets for modeling.
- Built and deployed an interactive **AI chatbot using Streamlit**, enabling real-time user interaction through a simple web interface.
- Collaborated on transforming theoretical ML knowledge into a **working end-to-end application**.

Projects

OptiScholar: Personalized Intelligent Multi-model Scholarship Recommendation System

02/2026 – 06/2026

Graduation Project | Python, PyTorch, TensorFlow, DL, NLP

- Developed an end-to-end AI-powered scholarship recommendation system combining eligibility filtering, personalized recommendation, document processing, and conversational guidance.
- Implemented and compared multiple recommendation and deep learning approaches, including **Neural Collaborative Filtering (NCF), BiLSTM, and SBERT-based components**.
- Built data pipelines covering preprocessing, interaction modeling, negative sampling, data sparsity, and class imbalance.
- Evaluated recommendation performance using **Precision@K, Recall@K, and Hit Rate** to compare model effectiveness.
- Designed system architecture integrating AI models with document parsing, recommendation, and chatbot components.

Medical Question Answering using LLM Fine-Tuning (QLoRA)

2026 – 2026

Python, PyTorch, Hugging Face, Unsloth, NLP, LLMs

- Fine-tuned **LLaMA 3.2 1B** for medical question answering using **Parameter-Efficient Fine-Tuning (PEFT) with QLoRA**.
- Applied 4-bit quantization and low-rank adaptation to reduce training cost while updating a parameter-efficient subset of model parameters.
- Preprocessed and structured a **16K+ medical Q&A dataset** into instruction-based dialogue format for supervised fine-tuning.
- Designed prompt engineering strategies for conversational inference and evaluated model behavior using **BERTScore, LLM-as-a-Judge, and semantic truthfulness metrics**.
- Achieved **7.88% improvement in BERTScore and 14% improvement in semantic truthfulness**, alongside a reduction in hallucination rate.

Comparative Study of Generative Models for Facial Image Synthesis

2026 – 2026

DL, Computer Vision, GAN, VAE, Diffusion models

- Implemented and evaluated **DCGAN, VQ-VAE, and Latent Diffusion Model (LDM)** approaches for facial image synthesis using the CelebA-HQ dataset.
- Compared models based on image quality, reconstruction performance, training stability, and computational cost.
- Evaluated generated images using **FID, SSIM, and PSNR** and performed large-scale image preprocessing and dataset preparation.

Secure InstaPay System – DevSecOps Implementation

2026 – 2026

DevSecOps, CI/CD, Docker, GitHub Actions, Security Testing

- Designed a secure CI/CD pipeline for an InstaPay-like application using **GitHub Actions and Docker**.
- Integrated automated build, testing, and security validation into the development pipeline.
- Implemented security testing scenarios covering input validation, authentication, data handling, and application behavior.
- Applied secure coding and deployment practices across frontend, backend, and deployment components.

Education

Bachelor of informatics and computer science - Artificial Intelligence

2022 – 2026

The British University in Egypt